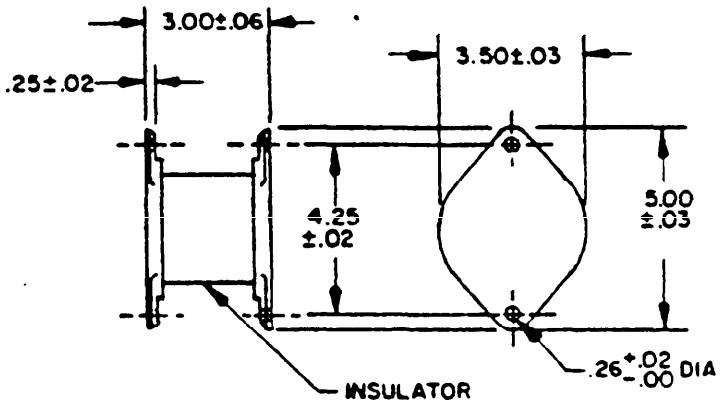
MILITARY SPECIFICATION SHEET
CAPACITORS, FIXED, MICA DIELECTRIC,
STYLES CM80, CM81, AND CM83

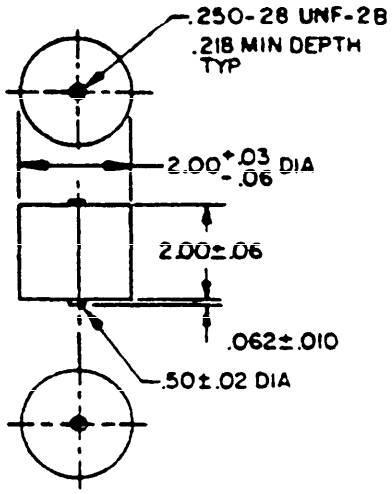
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.

STYLE CM80 IS INACTIVE FOR NEW DESIGN AFTER THE DATE OF THIS SPECIFICATI



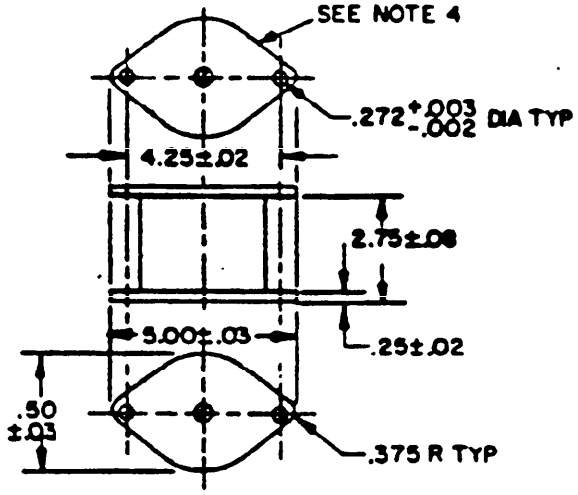
STYLE CM80

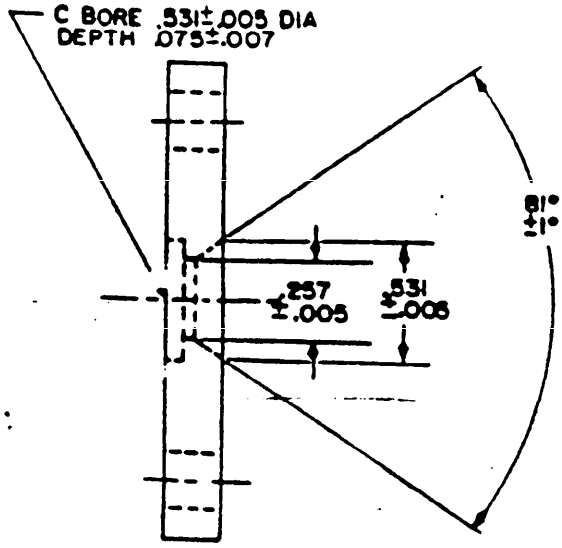


STYLE CM81

FS

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STYLE CM83

INCHES	MM
.002	.05
.003	.08
.005	.13
.010	.25
.02	.51
.03	.76
.06	1.52
.062	1.57
.08	2.03
.218	5.54
.25	6.35
.257	6.53
.26	6.60
.272	6.91
.375	9.53
.50	12.70
.531	13.49
2.50	63.50
2.75	69.85
3.00	76.20
3.50	88.90
4.25	107.95
5.00	127.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005(.13 mm) for three place decimals.
4. Brackets (flanges) shall be removable.
5. Hardware to be plated against corrosion.
6. Approximate weight CM85 is 5.5 pounds (2.49 kg).

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REQUIREMENTS:

Dimensions: See figure.

Material: Bracket (flange) aluminum per QQ-A-250/4 (T3-temper)

Construction: CM80 - Molded case. CM81 and CM83 - Cast epoxy case.

Capacitance value: See table I.

Capacitance tolerance: ±5% (symbol J).

Peak working voltage: See table I.

Operating temperature range:

Symbol	Temperature range
M	-55 to +70° C
O	-55 to +125° C

Vibration grade: Grade 1 (10-55 Hz).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/° C	Capacitance drift
B	Not rated	Not rated
D	-100 to +100	± (0.3 percent +0.1 pF)
E	-20 to +100	± (0.1 percent +0.1 pF)
F	0 to +70	± (0.05 percent +0.1 pF)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
AC sine-wave potential rms equal to peak working voltage at 100 cycles or less for not less than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202; condition B (50,000 ft).
Test potential: Capacitors rated above 6,000 volts - 60 percent of rated peak voltage; capacitors rated at 4,000 thru 6,000 volts - 100 percent of rated peak voltage; capacitors rated at 3,000 volts or less - 150 percent of rated peak voltage.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 25° C:
(CM80) - 7,500 megohms minimum.
(CM81 and CM83) - 10,000 megohms, minimum.
At 55° C:
(CM80) - 0 to 5,000 pF -- 2,000 megohms, minimum.
Greater than 5,000 pF -- 10 megohm-microfarads.
At 125° C:
(CM81 and CM83) - 0 to 5,000 pF -- 3,000 megohms, minimum. Greater than 5,000 pF -- 15 megohm-microfarads.

Dissipation factor, maximum - See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Current rating:
Passage of rated current shall not cause the temperature of the capacitor body's external surface to rise above the ambient temperature by more than 15° C for style CM80 and not more than 40° C for styles CM81 and CM83.

Vibration:
Method 201 of MIL-STD-202.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except the high temperature shall be as follows:
Symbol M - +70° C.
Symbol O - +125° C.

Immersion cycling:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - 3,000 megohms, minimum.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±4 percent of nominal value or 1 pF, whichever is greater.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G(pk)).

Moisture resistance:
Method 106 of MIL-STD-202:
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial value.

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Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±2 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.

Temperature coefficient and capacitance drift: Within the limits specified for the applicable characteristic.

Low temperature storage and life:
48 hours at -55 $^{+0°}_{-2}$ C immediately followed by qualification or performance test, as applicable.
Qualification test - 1,000 hours at the applicable high test temperature with rated current at 1 MHz
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±8 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±3 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.

Performance check - 250 hours at conditions specified for qualification test.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±2 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.

Continuation test - Units that have been subjected to the 250 hour test shall be continued for an additional 750 hours and meet the requirements for the qualification test. (This test not applicable to styles CM81 and CM83.)

Qualification inspection:
Qualification inspection (Group E) is required.

Supersession data: Styles CM82 (all characteristics), CM81 and CM83 (characteristics B, C, and F) have not been qualified; therefore, no interchangeability and substitution data are offered.

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TABLE I. Styles CM80, CM81, and CM83.

Type designation 1/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps)				Characteristic
			3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	
CM80-R470JM1	47	10,000	2.7	2.0	0.91	0.27	B, D, F
CM80-R510JM1	51	10,000	2.7	2.0	1.0	0.30	
CM80-R560JM1	56	10,000	3.0	2.2	1.0	0.30	
CM80-R620JM1	62	10,000	3.3	2.2	1.1	0.33	
CM80-R680JM1	68	10,000	3.3	2.4	1.2	0.33	
CM80-R750JM1	75	10,000	3.6	2.4	1.2	0.36	
CM80-R820JM1	82	10,000	3.6	2.7	1.3	0.39	
CM80-R910JM1	91	10,000	3.9	2.7	1.5	0.43	
CM80-R101JM1	100	10,000	4.3	3.0	1.5	0.47	
CM80-R111JM1	110	10,000	4.3	3.0	1.6	0.51	
CM80-R121JM1	120	10,000	4.7	3.3	1.6	0.56	
CM80-R131JM1	130	10,000	4.7	3.6	1.8	0.62	
CM80-R151JM1	150	10,000	5.1	3.6	1.8	0.62	
CM80-R161JM1	160	10,000	5.1	3.9	2.0	0.75	
CM80-R181JM1	180	10,000	5.6	4.3	2.2	0.82	
CM80-R201JM1	200	10,000	6.2	4.3	2.4	0.82	
CM80-R221JM1	220	10,000	6.2	4.7	2.4	0.91	
CM80-R241JM1	240	10,000	6.8	5.1	2.7	1.0	
CM80-R271JM1	270	10,000	6.8	5.1	2.7	1.0	
CM80-R301JM1	300	10,000	7.5	5.6	3.0	1.1	
CM80-R331JM1	330	10,000	7.5	5.6	3.0	1.2	
CM80-R361JM1	360	10,000	8.2	6.2	3.6	1.2	
CM80-R391JM1	390	10,000	8.2	6.2	3.6	1.3	
CM80-R431JM1	430	10,000	9.1	6.2	3.6	1.3	
CM80-R471JM1	470	10,000	9.1	6.8	3.9	1.5	
CM80-R511JM1	510	10,000	9.1	6.8	3.9	1.6	
CM80-R561JM1	560	10,000	10	7.5	4.3	1.6	
CM80-R621JM1	620	10,000	11	8.2	4.7	1.8	
CM80-R681JM1	680	10,000	11	8.2	4.7	1.8	
CM80-R751JM1	750	10,000	11	8.2	5.1	2.0	
CM80-R821JM1	820	10,000	12	9.1	5.1	2.2	
CM80-R911JM1	910	10,000	12	9.1	5.6	2.2	
CM80-R102JM1	1000	10,000	12	10	6.2	2.4	
CM80-R112JM1	1100	10,000	13	10	6.2	2.7	
CM80-R122JM1	1200	10,000	15	11	6.8	2.7	
CM80-R132JM1	1300	10,000	15	11	6.8	2.7	
CM80-R152JM1	1500	10,000	15	12	7.5	3.0	
CM80-R162JM1	1600	10,000	16	12	7.5	3.3	
CM80-R182JM1	1800	10,000	16	13	8.2	3.6	
CM80-R202JM1	2000	10,000	16	13	8.2	3.6	
CM80-Q222JM1	2200	8,000	18	13	9.1	3.9	
CM80-Q242JM1	2400	8,000	18	15	10	4.3	
CM80-Q272JM1	2700	8,000	18	15	10	4.3	
CM80-Q302JM1	3000	8,000	18	16	10	4.7	
CM80-Q332JM1	3300	8,000	20	18	11	4.7	
CM80-Q362JM1	3600	8,000	20	18	11	5.1	
CM80-Q392JM1	3900	8,000	20	18	12	5.6	
CM80-Q432JM1	4300	8,000	20	18	12	5.6	
CM80-P472JM1	4700	6,000	20	20	13	6.2	
CM80-P512JM1	5100	6,000	20	20	13	6.2	
CM80-N562JM1	5600	5,000	20	20	13	6.8	
CM80-N622JM1	6200	5,000	20	22	15	6.8	
CM80-N682JM1	6800	5,000	20	22	15	7.5	B, D, F

See footnotes at end of table.

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TABLE I. Styles CM80, CM81, and CM83. - Continued

Type designation 2/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM80-N752JM1	7500	5,000	20	22	15	7.5	B, D, F
CM80-N822JM1	8200	5,000	20	24	16	8.2	
CM80-N912JM1	9100	5,000	20	24	16	8.2	
CM80-N103JM1	10,000	5,000	20	24	18	9.1	
CM80-N113JM1	11,000	5,000	20	27	18	9.1	
CM80-M123JM1	12,000	4,000	20	27	18	10	
CM80-M133JM1	13,000	4,000	20	27	20	10	
CM80-M153JM1	15,000	4,000	20	27	20	11	
CM80-M163JM1	16,000	4,000	20	27	20	11	
CM80-L183JM1	18,000	3,000	20	30	20	12	
CM80-L203JM1	20,000	3,000	20	30	22	13	
CM80-L223JM1	22,000	3,000	20	30	22	13	
CM80-L243JM1	24,000	3,000	20	33	24	13	
CM80-L273JM1	27,000	3.000	20	33	24	13	
CM80-J303JM1	30,000	2,000	20	33	24	15	
CM80-J333JM1	33,000	2,000	20	33	24	15	
CM80-J363JM1	36,000	2,000	20	33	27	16	
CM80-J393JM1	39,000	2,000	20	36	27	16	
CM80-J433JM1	43,000	2,000	20	36	27	16	
CM80-J473JM1	47,000	2,000	20	36	27	18	
CM80-J513JM1	51,000	2,000	20	36	30	18	
CM80-J563JM1	56,000	2,000	20	36	30	18	
CM80-J623JM1	62,000	2,000	20	39	30	18	
CM80-J683JM1	68,000	2,000	20	39	33	20	
CM80-H753JM1	75,000	1,500	20	39	33	20	
CM80-H823JM1	82,000	1,500	20	39	33	20	
CM80-H913JM1	91,000	1,500	20	39	33	20	
CM80-H104JM1	100,000	1,500	20	39	36	20	B, D, F
CM8--R470JO1	47	10,000	2.7	2.0	.91	.27	D, E
CM8--R510JO1	51	10,000	2.7	2.0	1.0	.30	
CM8--R560JO1	56	10.000	3.0	2.2	1.0	.30	
CM8--R620JO1	62	10,000	3.3	2.2	1.1	.33	
CM8--R680JO1	68	10,000	3.3	2.4	1.2	.33	
CM8--R750JO1	75	10,000	3.6	2.4	1.2	.36	
CM8--R820JO1	82	10,000	3.6	2.7	1.3	.39	
CM8--R910JO1	91	10,000	3.9	2.7	1.5	.43	
CM8--R101JO1	100	10,000	4.3	3.0	1.5	.47	
CM8--R111JO1	110	10,000	4.3	3.0	1.6	.51	
CM8--R121JO1	120	10,000	4.7	3.3	1.6	.56	
CM8--R131JO1	130	10,000	4.7	3.6	1.8	.62	
CM8--R151JO1	150	10,000	5.1	3.6	1.8	.62	
CM8--R161JO1	160	10,000	5.1	3.9	2.0	.75	
CM8--R181JO1	180	10,000	5.6	4.3	2.2	.82	
CM8--R201JO1	200	10,000	6.2	4.3	2.4	.82	
CM8--R221JO1	220	10.000	6.2	4.7	2.4	.91	
CM8--R241JO1	240	10.000	6.8	5.1	2.7	1.0	
CM8--R251JO1	250	10,000	6.8	5.1	2.7	1.0	
CM8--R271JO1	270	10,000	6.8	5.1	2.7	1.0	
CM8--R301JO1	300	10,000	7.5	5.6	3.0	1.1	
CM8--R331JO1	330	10,000	7.5	5.6	3.0	1.2	
CM8--R361JO1	360	10,000	8.2	6.2	3.6	1.2	
CM8--R391JO1	390	10,000	8.2	6.2	3.6	1.3	
CM8--R431JO1	430	10,000	9.1	6.2	3.6	1.3	D, E

See footnote at end of table.

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TABLE I. Styles CM80, CM81, and CM83. - Continued

Type designation 2/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM8--R471JO1	470	10,000	9.1	6.8	3.9	1.5	D, E
CM8--R511JO1	510	10,000	9.1	6.8	3.9	1.6	
CM8--R561JO1	560	10,000	10	7.5	4.3	1.6	
CM8--R621JO1	620	10,000	11	8.2	4.7	1.8	
CM8--R681JO1	680	10,000	11	8.2	4.7	1.8	
CM8--R751JO1	750	10,000	11	8.2	5.1	2.0	
CM8--R821JO1	820	10,000	12	9.1	5.1	2.2	
CM8--R911JO1	910	10,000	12	9.1	5.6	2.2	
CM8--R102JO1	1000	10,000	12	10	6.2	2.4	
CM8--R112JO1	1100	10,000	13	10	6.2	2.7	
CM8--R122JO1	1200	10,000	15	11	6.8	2.7	
CM8--R132JO1	1300	10,000	15	11	6.8	2.7	
CM8--R152JO1	1500	10,000	15	12	7.5	3.0	
CM8--R162JO1	1600	10,000	16	12	7.5	3.3	
CM8--R182JO1	1800	10,000	16	13	8.2	3.6	
CM8--R202JO1	2000	10,000	16	13	8.2	3.6	
CM8--Q222JO1	2200	8,000	18	13	9.1	3.9	
CM8--Q242JO1	2400	8,000	18	15	10	4.3	
CM8--Q272JO1	2700	8,000	18	15	10	4.3	
CM8--Q302JO1	3000	8,000	18	16	10	4.7	
CM8--Q332JO1	3300	8,000	20	16	11	4.7	
CM8--Q362JO1	3600	8,000	20	18	11	5.1	
CM8--Q392JO1	3900	8,000	20	18	12	5.6	
CM8--Q432JO1	4300	8,000	20	18	12	5.6	
CM8--P472JO1	4700	6,000	20	20	13	6.2	
CM8--P512JO1	5100	6,000	20	20	13	5.2	
CM8--N562JO1	5600	5,000	20	20	13	6.8	
CM8--N622JO1	6200	5,000	20	22	15	6.8	
CM8--N682JO1	6800	5,000	20	22	15	7.5	
CM8--N752JO1	7500	5,000	20	22	15	7.5	
CM8--N822JO1	8200	5,000	20	24	16	8.2	
CM8--N912JO1	9100	5,000	20	24	16	8.2	
CM8--N103JO1	10,000	5,000	20	24	18	9.1	
CM8--N113JO1	11,000	5,000	20	27	18	9.1	
CM8--M123JO1	12,000	4,000	20	27	18	10	
CM8--M133JO1	13,000	4,000	20	27	20	10	
CM8--M153JO1	15,000	4,000	20	27	20	11	
CM8--M163JO1	16,000	4,000	20	27	20	11	
CM8--L183JO1	18,000	3,000	20	30	20	12	
CM8--L203JO1	20,000	3,000	20	30	22	13	
CM8--L223JO1	22,000	3,000	20	30	22	13	
CM8--L243JO1	24,000	3,000	20	33	24	13	
CM8--L273JO1	27,000	3,000	20	33	24	13	
CM8--J303JO1	30,000	2,000	20	33	24	15	
CM8--J333JO1	33,000	2,000	20	33	24	15	
CM8--J363JO1	36,000	2,000	20	33	27	16	
CM8--J393JO1	39,000	2,000	20	36	27	16	
CM8--J433JO1	43,000	2,000	20	36	27	16	
CM8--J473JO1	47,000	2,000	20	36	27	18	
CM8--J513JO1	51,000	2,000	20	36	30	18	
CM8--J563JO1	56,000	2,000	20	36	30	18	
CM8--J623JO1	62,000	2,000	20	39	30	18	
CM8--J683JO1	68,000	2,000	20	39	33	20	
CM8--H753JO1	75,000	1,500	20	39	33	20	
CM8--H823JO1	82,000	1,500	20	39	33	20	
CM8--H913JO1	91,000	1,500	20	39	33	20	
CM8--H104JO1	100,000	1,500	20	39	36	20	D, E

1/ For style CM80, the complete type designation will include an additional symbol to indicate characteristic.
2/ For styles CM81 and CM83, the complete type designation will include additional symbols to indicate style and characteristic.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy - AS, OS, EC, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - MC, CG
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 REPLACES EDITION OF JAN 88 WHICH MAY BE USED

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